

AI Principles and Practice

Integrated 12-Week Course Structure

Fundamentals first, frontier earned

Module COMP-0987 · Level 9 (Postgraduate) · 10 Credits
Department of Computing and Mathematics

1. The Approach

How fundamentals pay off later (the spine)

Taught earlier	Pays off later as
Data types, distributions, EDA (Wk 1)	Data-centric AI and augmentation (Wk 9)
Train/test, overfitting, evaluation (Wk 2)	Honest benchmarking of foundation models (Wk 9)
Bias and fairness (Wk 3)	The fairness audit in the final project
Safety, governance and explainability (Wk 4)	The model card and governance note in the final project
PCA and embeddings (Wk 5)	Representation learning and where bias hides (Wk 10)
Neuron to MLP, backprop, training dynamics (Wk 7)	Every deep model in the second half; neural ODEs (Wk 11)
SVD and low-rank structure (Wk 8)	LoRA and parameter-efficient fine-tuning (Wk 9)
Cross-entropy as information (Wk 8)	Diffusion and generative objectives (Wk 11)

Teaching team

Lecturer	Weeks delivered
Indrakshi	W1 Data Analysis · W7 Neural Networks · W8 Mathematics · W9 Transfer Learning · W11 Diffusion, ODEs & Symmetry
Jonathan	W2 Supervised Learning · W3 Ethics, Bias & Fairness · W4 AI Safety, Governance & XAI · W5 Unsupervised · W10 Transformers

Weeks 6 (quiz and reflection) and 12 (final-project delivery) are run jointly by the teaching team.

2. Learning Outcome Mapping

LO	Description	Primary weeks
LO1	Apply mathematical principles (linear algebra, probability, calculus)	7, 8, 11
LO2	Evaluate and distinguish ML paradigms; select appropriate methods	2, 5, 9
LO3	Design / implement DL models with industry tools; visualise metrics and limitations	2, 7, 10, 11
LO4	Assess structure and learning behaviour of transformer architectures	10
LO5	Insight in ethics and safety: fairness, explainability, data governance	1, 3, 4

3. Weekly Plan

Phase	Span
Phase I — Foundations and Ethics	Weeks 1 to 5
Phase II — Neural Networks and the Mathematics	Weeks 7 and 8
Phase III — Modern and Science-Inspired Learning	Weeks 9 to 11
Milestones	Quiz and reflection (Week 6) · Final-project delivery (Week 12)

Week 1: Data Analysis and the ML Workflow

Phase I · LOs: LO2, LO5 · Lecturer: Indrakshi

Lecture (90 min)

- The end-to-end ML workflow; where everything in the course fits.
- Data analysis fundamentals: data types (discrete, continuous, categorical, ordinal, binary), distributions, summary statistics, correlation.
- Exploratory data analysis (EDA) and visualisation as a discipline.
- Data processing: batch versus stream, missing-data imputation, outlier detection, leakage-safe pipelines.
- Data governance and dataset documentation introduced now.

Lab: Know Your Data

- Full EDA on a chosen project dataset: profile types, distributions, missingness, outliers.
- Build a reproducible preprocessing pipeline (scikit-learn).
- Produce a data dictionary and datasheet.

Week 2: Supervised Learning: Regression, Classification and Practical Applications

Phase I · LOs: LO2, LO3 · Lecturer: Jonathan

Lecture (90 min)

- The supervised setup: features, targets, train / validation / test.
- Regression (linear, polynomial, regularised) and classification (logistic regression, k-NN, trees, ensembles).
- Loss functions; overfitting, underfitting and the bias–variance trade-off.
- Evaluation done properly: cross-validation, and metrics with visualisation (RMSE / R^2 , precision / recall / F1, confusion matrix, ROC / AUC).
- **Practical applications:** framing real problems as regression or classification — demand forecasting and pricing, credit scoring, churn prediction, medical diagnosis, fraud and spam detection, predictive maintenance — and how the choice of target and metric follows from the business or scientific question.

Lab: Build, Evaluate, Compare

- Train regression and classification models on the project data.
- Diagnose over- and underfitting with learning curves.

- Produce a visual evaluation report; reason about which metric matters for the chosen application and why accuracy can mislead.

Week 3: Ethics, Bias and Fairness

Phase I · LOs: LO5, LO3 · Lecturer: Jonathan

Lecture (90 min)

- Foundations of AI ethics; why fairness is a modelling concern, not an afterthought.
- Types of bias in AI systems: data, algorithmic and societal; where bias enters the pipeline.
- Fairness definitions and metrics: group fairness, demographic parity, equalised odds, disparate impact — building directly on the Week 2 evaluation metrics.
- The unavoidable trade-offs between fairness criteria (impossibility results) and what “algorithmic justice” can and cannot deliver.

Lab: Auditing a Model for Bias

- Audit the Week 2 and project models with Fairlearn or AIF360; measure group metrics.
- Attempt a mitigation and quantify its cost in accuracy and other metrics.
- Write a short ethical-impact assessment.

Week 4: AI Safety, Governance and Explainability (XAI)

Phase I · LOs: LO5, LO3 · Lecturer: Jonathan

Lecture (90 min)

- **AI safety:** specification and alignment, robustness and distribution shift, failure modes (specification gaming, reward hacking, silent failure), and the safety-versus-capability distinction across the model lifecycle.
- **Governance and regulation:** GDPR and the EU AI Act risk tiers; risk-management frameworks (e.g. the NIST AI RMF); data governance, accountability and audit trails.
- **Documentation as governance:** model cards, datasheets and system cards as the artefacts that make oversight possible.
- **Explainability (XAI):** post-hoc methods — SHAP, LIME, feature importance and surrogate models — applied to the classical models from Week 2, together with the limits of interpretability and where explanations mislead.

Lab: Explain and Govern

- Apply SHAP and LIME to the Week 2 and project models; reason about which features drive decisions.
- Probe robustness: measure performance under a deliberate distribution shift.
- Produce a model card and a short governance note for the project model.

Week 5: Unsupervised Learning and the Idea of a Representation

Phase I · LOs: LO1, LO2 · Lecturer: Jonathan

Lecture (90 min)

- Learning without labels: clustering (K-means, hierarchical, GMM) and choosing K.
- Dimensionality reduction and PCA — the first formal encounter with a learned representation.
- Embeddings introduced: data become vectors in a space where geometry carries meaning, the concept that unifies the whole course.
- How representations can encode the biases discussed in Weeks 3 and 4; cluster validity and the limits of unsupervised evaluation.

Lab: Finding Structure

- Cluster the project data; validate with silhouette and elbow.
- Apply PCA; visualise the data manifold in 2D.
- Interpret what the structure means in the domain.

Week 6: Quiz and Reflection Milestone

Milestone · No new teaching content

An in-class quiz on Weeks 1 to 5 (data analysis; supervised learning; ethics, bias and fairness; AI safety, governance and explainability; and unsupervised learning), followed by consolidation, feedback on the first assignment, and a project check-in.

Week 7: Neural Networks and Deep Learning: Introduction

Phase II · LOs: LO1, LO3 · Lecturer: Indrakshi

Lecture (90 min)

- From a single neuron to the multi-layer perceptron; activation functions; the forward pass and loss.

- Backpropagation derived properly (the chain rule made concrete); the training loop: epochs, batches, gradient descent.
- Training well: optimisers (SGD, momentum, Adam) and learning-rate schedules; regularisation (weight decay, dropout, early stopping, batch and layer norm).
- Training dynamics: healthy versus pathological learning; vanishing and exploding gradients; gradient descent as gradient flow (the dynamical-systems view that returns in Week 11).

Lab: Build It, Then Train It Well

- Implement a tiny MLP and backprop from scratch in NumPy; reproduce it in PyTorch and verify the gradients match.
- Train on the project data; plot loss and accuracy curves.
- Systematically improve it (optimiser, regularisation, schedule); diagnose and fix a deliberately broken run.

Week 8: The Mathematics That Matters

Phase II · LOs: LO1 · Lecturer: Indrakshi

Lecture (90 min)

- Linear algebra deepened: eigendecomposition and SVD; low-rank structure, with a preview that this is what makes LoRA work (Week 9).
- Information theory: entropy, cross-entropy and KL divergence, showing why the loss you have been minimising is principled, not arbitrary; mutual information and the information-bottleneck intuition.
- The manifold hypothesis and high-dimensional geometry: why real data is low-dimensional and distances behave strangely.

Lab: The Math Behind the Models

- Implement PCA via SVD from scratch; connect to low-rank approximation.
- Compute entropy, cross-entropy and KL on real model outputs; tie directly to the training loss.
- Estimate the intrinsic dimensionality of the project-data embeddings.

Week 9: Transfer Learning, Foundation Models and Data Augmentation

Phase III · LOs: LO2, LO3 · Lecturer: Indrakshi

Lecture (90 min)

- Why training from scratch is increasingly the exception; foundation models, pretraining and the self-supervised objectives that produce them.
- The transfer spectrum: zero-shot to prompting to linear probing to fine-tuning to PEFT and LoRA (the payoff of the low-rank maths in Week 8).
- Data augmentation as encoding invariances; synthetic data, with its uses and risks.
- Cost, compute and carbon as model-selection criteria.

Lab: Adapt, Do Not Rebuild

- Adapt a pretrained backbone three ways — linear probe, full fine-tune and LoRA; compare accuracy, trainable parameters and time.
- Improve results through augmentation alone (data-centric experiment).

Week 10: Transformers and Attention

Phase III · LOs: LO3, LO4 · Lecturer: Jonathan

Lecture (90 min)

- Self-attention as content-based retrieval; multi-head attention; positional encoding; encoder and decoder blocks.
- Learning behaviour: in-context learning, emergence, and scaling laws as an empirical science.
- Symmetry and inductive bias: how transformers relate to CNNs and GNNs.
- Strengths, failure modes and compute realities.

Lab: Inside the Transformer

- Use a pretrained transformer for a task; extract and visualise attention.
- Probe intermediate representations (supports LO4).
- Run a small in-context-learning demonstration.

Week 11: Science-Inspired Learning: Diffusion, Neural ODEs and Symmetry

Phase III · LOs: LO1, LO3 · Lecturer: Indrakshi

Lecture (90 min)

- Diffusion models from non-equilibrium thermodynamics: add noise, then learn to reverse it; score functions.

- Neural ODEs: continuous-depth networks, the direct payoff of the gradient-flow view in Week 7.
- A brief look at energy-based models and geometric deep learning (symmetry and equivariance) as design patterns.
- The science-inspired recipe: borrow a mechanism from nature, make it learnable.

Lab: Generating with Physics

- Implement a minimal diffusion process; visualise the forward-noising and reverse-denoising trajectory.
- Generate samples and inspect the learned manifold.

Week 12: Final Project Delivery Milestone

Milestone · LOs: LO3, LO4, LO5

Students deliver, demonstrate and defend their final project in person. See the Final Project section for the full brief, deliverables and marking.

4. Final Project: End-to-End AI on an Unseen Dataset, with Live Defence

The capstone is a single, individual, end-to-end project that runs on a dataset the student has not seen before and that general-purpose AI assistants have not been trained on. It is assessed mainly on understanding, reproducibility and a live defence, so that producing the artefacts with a large language model is neither sufficient nor low-risk.

Deliverables

- A reproducible code repository with a documented commit history.
- A short logbook recording decisions, experiments, dead ends and fixes.
- A trained model with real loss and metric curves on the assigned dataset.
- From-scratch implementations of backpropagation, an optimiser step, and PCA via SVD, tested against references.
- An ethics and fairness audit of the student's own model on the assigned data: measured group metrics, one mitigation, and the measured cost of that mitigation.
- An explainability summary (SHAP or LIME) identifying the features that drive the model's decisions.
- A model card and a short governance note.
- A live demonstration and viva.

Marking (within the 40% final-project weight)

Element	Share of project
Live demonstration and viva	40%
Reproducibility and correctness of results	25%
From-scratch components and their tests	15%
Ethics, fairness and explainability audit	10%
Process artefacts (repository history, logbook)	10%

The final project maps to LO3, LO4 and LO5.

5. Assessment Overview

Component	Weight	LOs	Timing
Lab portfolio (continuous project notebooks)	25%	LO1 to LO3	Continuous
Quiz	15%	LO1, LO2	Week 6
Applied assignment: modelling with an ethics, fairness and explainability audit	20%	LO2, LO3, LO5	Around Week 7

Component	Weight	LOs	Timing
Final project and live defence: end-to-end AI on an unseen dataset	40%	LO3, LO4, LO5	Week 12

Indicative; tune to programme and QA requirements.

6. Balance and Delivery Notes

Phase	Weeks	Share	Covers
Foundations and ethics	1 to 5	50%	Data and statistics; supervised learning and applications; ethics, bias and fairness; AI safety, governance and explainability; unsupervised learning
Neural networks and the mathematics	7 and 8	20%	Build and train; SVD; information theory; manifolds
Modern and science-inspired	9 to 11	30%	Transfer and augmentation; transformers; diffusion and ODEs
Milestones	6 and 12	n/a	Quiz and reflection; final-project delivery

- Each teaching week is a 90-minute lecture plus a lab. There are 10 teaching weeks (1 to 5 and 7 to 11), split evenly between the two lecturers — five each.
- Ethics, bias and fairness sit at Week 3 on the classical models; AI safety, governance and explainability follow at Week 4, and both are applied again in the final-project fairness audit and governance note. Week 5 then notes how representations can carry the same biases.
- Week 7 is dense by design: neural networks are built and trained in one week, so the lab does the from-scratch build while the lecture also covers training well. If the cohort needs room, the lab can run across the Week 7 and Week 8 sessions.
- The mathematics in Week 8 consolidates tools used in Weeks 5 and 7 and sets up the transfer-learning week.
- Self-supervised pretraining is folded into Week 9 (Transfer Learning) as the objective behind foundation models, rather than taught as a separate week.

Infrastructure baseline. BYOD computer lab. Local workstations comparable to an x86 i7 CPU with 32 GB RAM and an NVIDIA GeForce RTX 3060 (12 GB) GPU.